Straight centroid loci for polar Poncelet polygons

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Abstract

For every size of the focus-circle Poncelet family inscribed in a parabola, the vertex and area centroids of the polar polygons trace complete lines parallel to the directrix. The proof removes every possible pole of their horizontal coordinates on the elliptic flag curve. In particular, finite zero-area configurations are accounted for rather than omitted.

Task 1: the original problem

Let

$$\mathcal{P}:\ y^2+4fx=0, \qquad \mathcal{C}:\ (x+f)^2+y^2=r^2, \qquad 0<r<f,$$

admit a primitive Poncelet family of $N\geq 3$ distinct vertices circumscribed about the inner circle. These are the families studied in [1], including all its convex cases. For a polygon P in the family, let Q be the polygon whose sides are the tangents to $\mathcal{P}$ at the vertices of P. Area and area centroid use signed area and winding multiplicity for a self-intersecting polygon, as in [2, 3].

Theorem 1. *There is a real constant h_0 such that*

$$\{C_0(Q) : P \text{ finite}\} = \{(h_0, Y) : Y\in\mathbb{R}\},$$
$$\{C_2(Q) : P \text{ finite}\} = \{(h_2, Y) : Y\in\mathbb{R}\}, \qquad h_2=\frac{8f^2-r^2}{12f}.$$

Thus both centroid loci are full lines parallel to the directrix $x=f$. Moreover, for each polygon individually,

$$A(Q)=-\tfrac{1}{2}A(P), \qquad (C_0(Q))_y=(C_0(P))_y, \qquad (C_2(Q))_y=(C_2(P))_y.$$

This proves Conjecture 5 of [1]. Its $N=3,4$ cases are already proved there. We use the standard elliptic-curve description of the Poncelet map from [2], and adapt the area zero-counting idea of [3]. The pole orders needed in this parabolic setting are established explicitly below. We do not infer a line from a generic conic-locus assertion.

Coordinates and polygon identities

Write the vertices of P as

$$P_i=(-t_i^2/(4f), t_i), \qquad i \pmod N.$$

The tangent at parameter t is $4fx+2ty-t^2=0$. Hence

$$Q_i=\left(-\frac{t_it_{i+1}}{4f}, \frac{t_i+t_{i+1}}{2}\right). \tag{1}$$

Put

$$S_2 = \sum_i t_i^2(t_{i+1} - t_{i-1}), \quad S_3 = \sum_i t_i^3(t_{i+1} - t_{i-1}), \quad S_{32} = \sum_i t_i^3(t_{i+1}^2 - t_{i-1}^2).$$

For any cyclic list of parameters, the shoelace formulas give

$$A(P) = -\frac{S_2}{8f}, \qquad A(Q) = \frac{S_2}{16f}, \tag{2}$$

$$(C_2(P))_y = (C_2(Q))_y = \frac{S_3}{3S_2}, \qquad (C_2(Q))_x = -\frac{S_{32}}{12fS_2}, \tag{3}$$

$$(C_0(Q))_y = \frac{1}{N}\sum_i t_i, \qquad (C_0(Q))_x = -\frac{1}{4fN}\sum_i t_i t_{i+1}. \tag{4}$$

These are polynomial identities for area and first moments before division by area. For example,

$$\det(Q_i, Q_{i+1}) = \frac{t_{i+1}^2(t_{i+2} - t_i)}{8f}.$$

Summing and shifting the index proves (2); multiplication by $(Q_i + Q_{i+1})/6$ gives the first moments in (3). All real finite polygons considered here have $A(P) \neq 0$: orient the edges with the circle on the left, and their fan triangles from the focus have area $r\ell_i/2 > 0$. Consequently $A(Q) \neq 0$ too.

The chord between parameters t, s has equation $4fx + (t+s)y - ts = 0$. Tangency to $\mathcal{C}$ is equivalent to

$$(ts + 4f^2)^2 = r^2\big(16f^2 + (t+s)^2\big). \tag{5}$$

If u, v are the two neighbors of t, Vieta's formula therefore gives

$$u + v = \frac{2(r^2 - 4f^2)t}{t^2 - r^2}. \tag{6}$$

These formulas also show that the polar vertices lie on $(x-f)^2/r^2 - y^2/(4f^2) = 1$.

The flag curve and the area zeros

Recall the standard complex flag curve

$$E = \{(p, \ell) : p \in \mathcal{P},\ p \in \ell,\ \ell \text{ tangent to } \mathcal{C}\}.$$

Its compactification is a smooth elliptic curve. In the coordinate above it has a model

$$w^2 = t^4 + 8f^2t^2 + 16f^2(f^2 - r^2).$$

The four roots on the right are distinct when $0 < r < f$, so the degree-two map to the t-sphere has four simple branch points. The two points over $t = \infty$ are unramified; t has a simple pole at each. The Poncelet step T is a translation of exact order N on E. The involution τ changing the chosen tangent reverses a polygon and satisfies $\tau T \tau = T^{-1}$. With a fixed point of τ as origin, these maps have the form $T(z) = z + a$, $\tau(z) = -z$; see [2, Section 3].

There are exactly $2N$ flags at which one of the t_i is infinite, and at each such flag exactly one t_i is infinite. To check that these are distinct, use the real geometry: the two real components of E correspond to the two choices of direction around the nested pair, and each projects bijectively to the real projective parabola. The step preserves each component and has N distinct vertices per orbit. The two flags over infinity belong to different components.

Near any such flag put the unbounded parameter equal to t. Its neighbors approach $r, -r$; all other parameters stay bounded. Splitting P into the triangle at that vertex and a bounded remainder shows

$$A(P) = \pm\frac{r}{4f}t^2 + o(t^2). \tag{7}$$

Thus $A(P)$, viewed as a meromorphic function on E, has exactly $2N$ poles, each of order two, and total pole order $4N$.

Now consider the N reflections $\rho_j = T^j\tau$, $0 \le j < N$. Each is an elliptic involution $z \mapsto ja - z$, and so has four fixed points (the four solutions of $2z = ja$). Their fixed-point sets are disjoint: a common fixed point for two reflections would be fixed by a nontrivial translation T^{j-k}. None of these $4N$ points is an infinite-vertex flag, since its real polygon has distinct vertices and cannot equal its reversed labeled vertex list up to a cyclic shift when $N \ge 3$.

Oriented area is invariant under T and changes sign under τ. It must therefore vanish at all $4N$ reflection fixed points. A nonconstant meromorphic function on a compact curve has the same total zero and pole order. By (7), these are precisely all zeros of $A(P)$, and every one is simple. Equation (2) gives exactly the same zero divisor for $A(Q)$.

Let $M_x(Q) = \frac{1}{6}\sum_i \det(Q_i, Q_{i+1})((Q_i)_x + (Q_{i+1})_x)$ be the signed first moment. It is finite where all t_i are finite, and it too is invariant under cyclic relabeling and changes sign under reversal. At each of the $4N$ zeros just found, $M_x(Q)$ vanishes. Since the denominator has a simple zero, the quotient $M_x(Q)/A(Q) = (C_2(Q))_x$ has a removable singularity there. There are no other possible finite poles. This supplies the finite-zero check directly, following the zero-counting method in [3] rather than assuming that a centroid's denominator never vanishes over $\mathbb{C}$.

The horizontal coordinates have no poles

First consider $(C_0(Q))_x$. In the sum in (4), the only unbounded terms near a pole parameter t combine as $t(u+v)$. Equation (6) makes this bounded. All other summands are bounded. Therefore $(C_0(Q))_x$ has no poles on E and is constant.

For $(C_2(Q))_x$, finite denominator zeros have already been removed. At an infinite-vertex flag, (7) and (2) give S_2 a nonzero term of order t^2. In S_{32}, the summand with unbounded central parameter is

$$t^3(v^2 - u^2) = t^3(v - u)(v + u) = O(t^2),$$

because u, v are bounded and (6) gives $u+v = O(t^{-1})$. Each of the other summands is $O(t^2)$ as well. Thus the quotient in (3) stays bounded. Meromorphicity makes these singularities removable too. A holomorphic function on the compact connected curve E is constant, proving $(C_2(Q))_x = h_2$. The two constants are real, as the family contains real polygons.

The area-centroid constant can be evaluated at infinity. Label the neighbors so that $u \to r$, $v \to -r$. Then

$$\frac{S_2}{t^2} \longrightarrow -2r, \qquad \frac{S_{32}}{t^2} \longrightarrow -4r(r^2 - 4f^2) + r^3 - (-r)^3 = 2r(8f^2 - r^2).$$

In the second limit the central summand uses (6), the two neighboring summands contribute r^3 each, and all remaining summands vanish after division by t^2. Equation (3) therefore gives $h_2 = (8f^2 - r^2)/(12f)$.

Finally, the finite loci are complete lines. The unlabeled real polygons form a circle, the quotient by T of either real component of E. Removing its unique infinite-vertex orbit leaves an interval.

For $t \to \pm\infty$, the original centroids satisfy

$$(C_0(P))_y = t/N + O(1), \qquad (C_2(P))_y = t/3 + O(1).$$

The second expansion follows from the same dominant triangle used in (7). Equations (3)–(4) give the identical vertical coordinates for Q. They are continuous on the finite interval and have opposite infinite limits at its ends. Every real vertical coordinate is attained, completing the proof.

Verification and four proposed follow-up tasks

The verifier checks the tangency and Vieta formulas, exact polygon area/moment identities, pole cancellation, and numerical closed families. The compact-curve argument is the proof for arbitrary N; finite calculations are supplementary checks. The source's perpendicular/directrix typo in its separate pentagon conjecture is not used, and the perimeter-centroid degree is not asserted here.

1. **Task 2.** Express h_0 in terms of N, r/f, and explicit elliptic-function invariants of the Poncelet map.
2. **Task 3.** Determine the directions and offsets of the two polar centroid lines when the circular caustic center is displaced off the parabola's axis.
3. **Task 4.** Determine the algebraic degree of the polar perimeter-centroid locus for every N, including the pentagon case.
4. **Task 5.** Classify inner conics of a parabola for which the vertex and area centroids of the polar polygons trace parallel lines for every available finite Poncelet period.

References

[1] F. Bellio, R. Garcia, D. Reznik, *Parabola-Inscribed Poncelet Polygons Derived from the Bicentric Family*, KoG **26** (2022), 3–15, doi:10.31896/k.26.1; arXiv:2111.00979v3.

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[3] A. C. Chavez-Caliz, *More about areas and centers of Poncelet polygons*, Arnold Math. J. **7** (2021), 91–106, doi:10.1007/s40598-020-00154-8; arXiv:2004.05404.